

Noise sources in PTAs and how to model them



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Visual by U.S. National Science Foundation

Summary



Credit: Abhimanyu Susobhanan

What is noise?

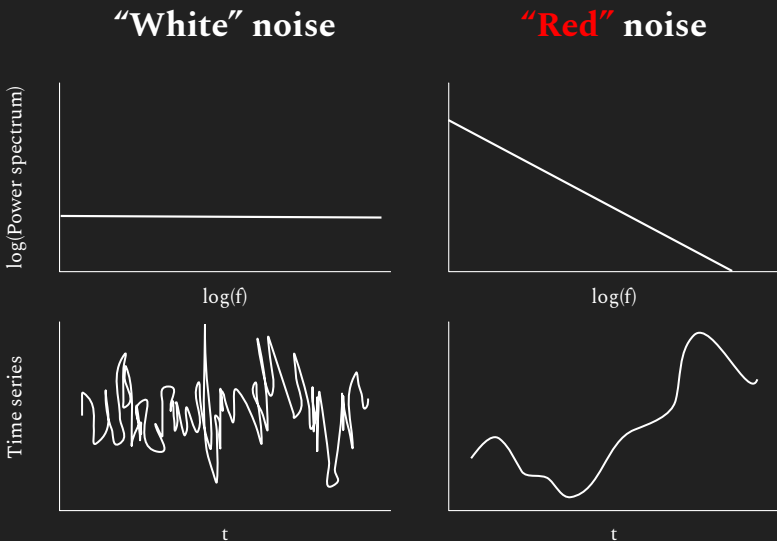
There are 2 definitions (that I can think of)

1. An unresolved, stochastic process, often defined by its spectrum
2. Anything that should be accounted for in your analysis, *besides your signal*.

What is noise?

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“The gravitational wave background is a common cross-correlated red noise process”

What is noise?

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2. Anything that should be accounted for in your analysis, *besides your signal*.

$$\begin{array}{ccc} \text{Data} & & \text{Model} \\ \underbrace{} & & \underbrace{} \\ \delta t = s + n \\ \begin{array}{ccc} \nearrow & & \nwarrow \\ \text{Signal} & & \text{Noise} \end{array} \end{array}$$

What is noise?

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1. An unresolved, stochastic process, often defined by its spectrum
2. Anything that should be accounted for in your analysis, *besides your signal*.

$$\delta t = \underbrace{M\epsilon}_{\text{Timing model}} + \underbrace{F_{\text{IRN}}a_{\text{IRN}}}_{\text{Intrinsic red noise}} + \boxed{F_{\text{GWB}}a_{\text{GWB}}} + \underbrace{n}_{\text{White noise}}$$

GWB (Signal)

What is noise?

There are 2 definitions (that I can think of)

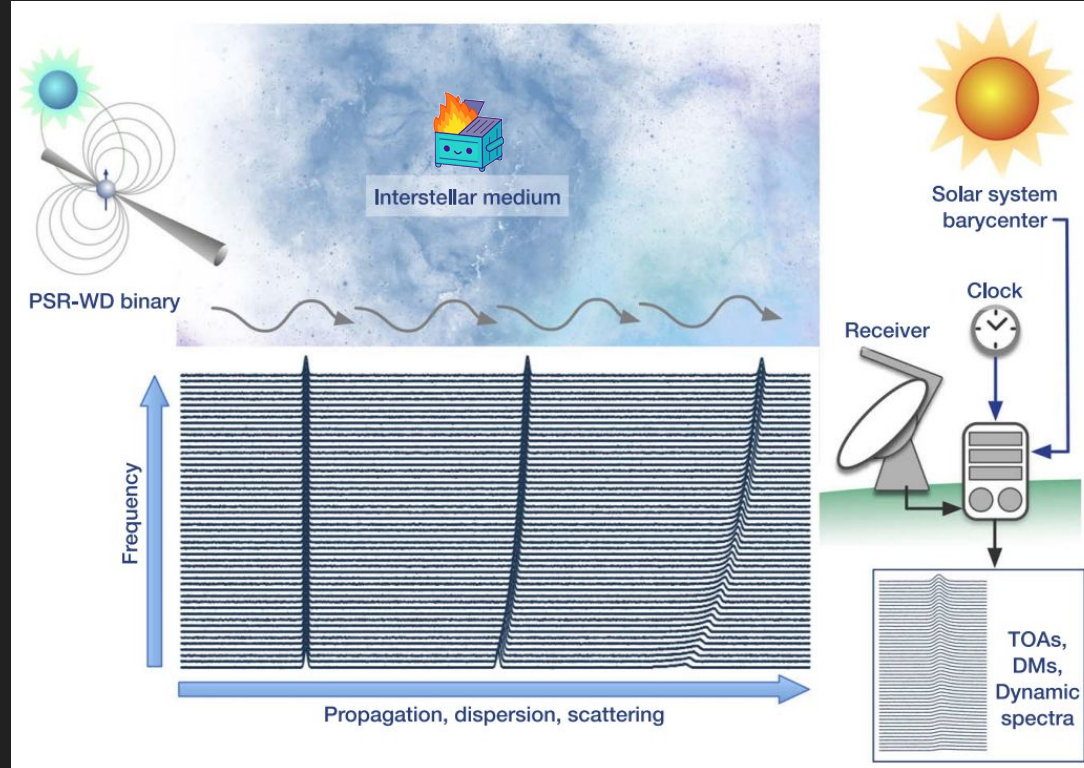
1. An unresolved, stochastic process, often defined by its spectrum
2. Anything that should be accounted for in your analysis, *besides your signal*.

$$\delta t = \boxed{M\epsilon} + \frac{F_{\text{IRN}} a_{\text{IRN}}}{\text{}} + \frac{F_{\text{GWB}} a_{\text{GWB}}}{\text{}} + \frac{n}{\text{}}$$

Timing model (signal) Intrinsic red noise GWB (noise) White noise

The goal of noise modeling is to adequately account for everything that can interfere with the measurement of our signal

A PTA noise budget

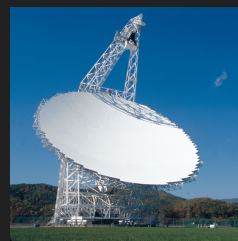
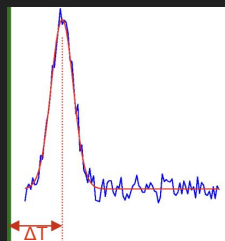
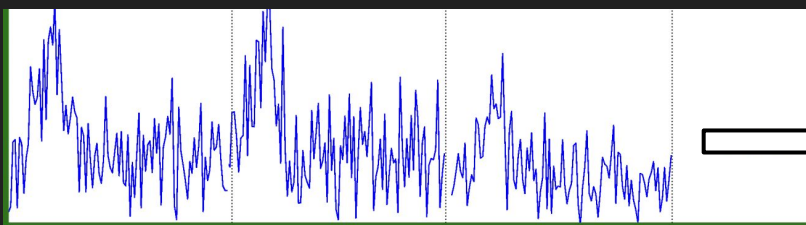


Agazie+2023: Detector Characterization and Noise Budget

White

Noise sources

A PTA noise budget



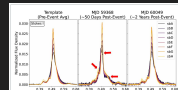
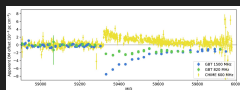
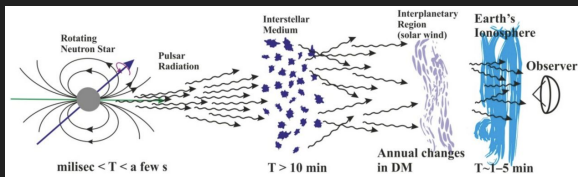
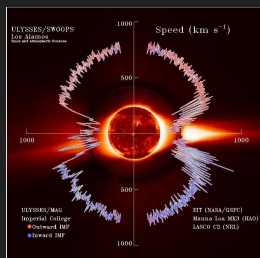
Red (time-correlated)

Chromatic:

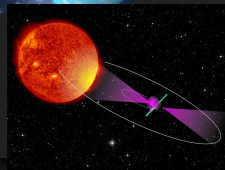
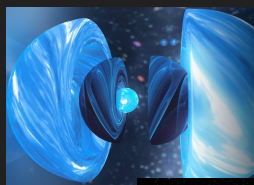


$$DM(t) = \int_{r_E}^{r_\star} n_e(\vec{r}, t) dl$$

$$\Delta t_{DM}(t) = \frac{e^2}{2\pi m_e c^2} \frac{DM(t)}{\nu^2}$$

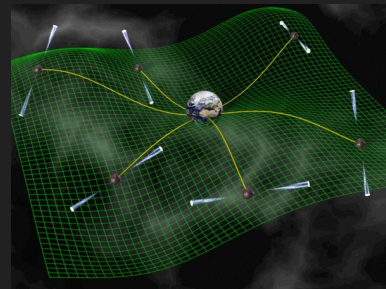


Intrinsic:



Achromatic:

Common:

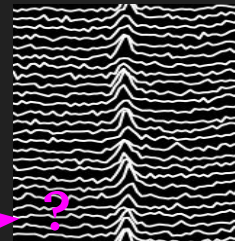
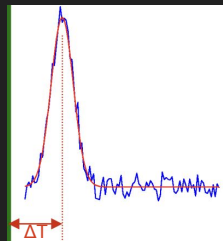
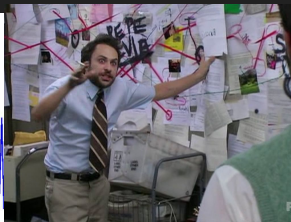
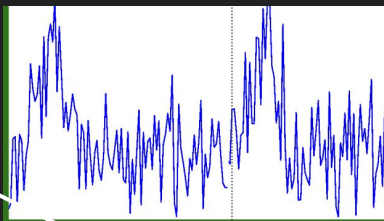


Noise sources

A PTA noise budget

White

Red (time-correlated)

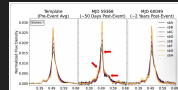
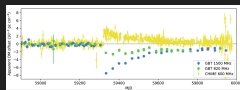
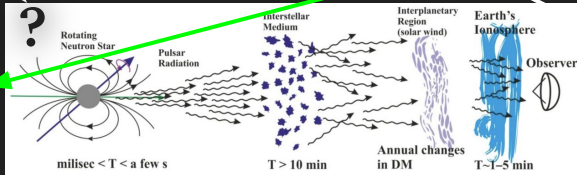
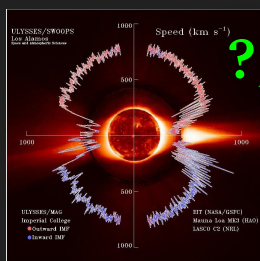


Chromatic:

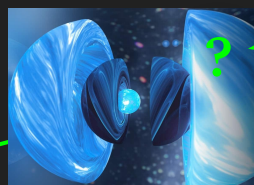


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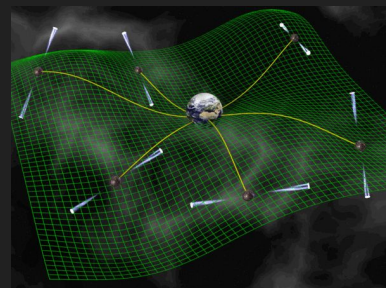
Intrinsic:



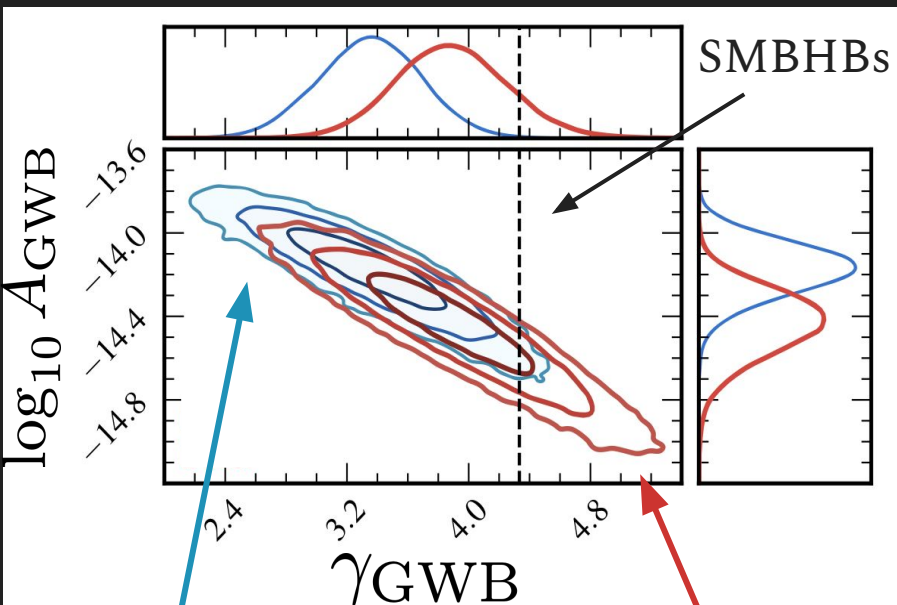
Achromatic:



Common:



Problem: Noise affects the measured GWB spectrum

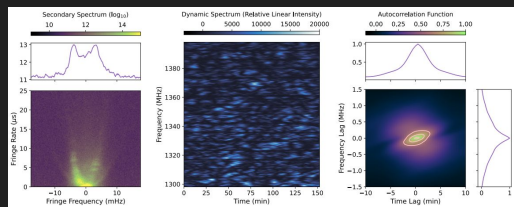
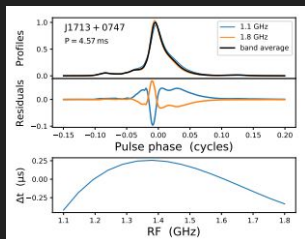


**Standard DM
model
(DMX)**

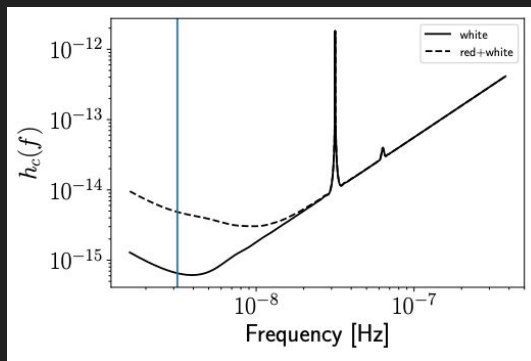
**alternative DM
model (DMGP)**

**What are these models and
which should we be using?**

Fundamental physics modeling

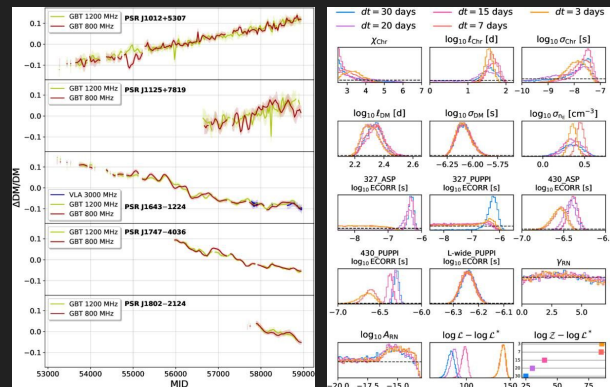


Impacts on PTA sensitivity



Agazie+2023: Detector Characterization and Noise Budget

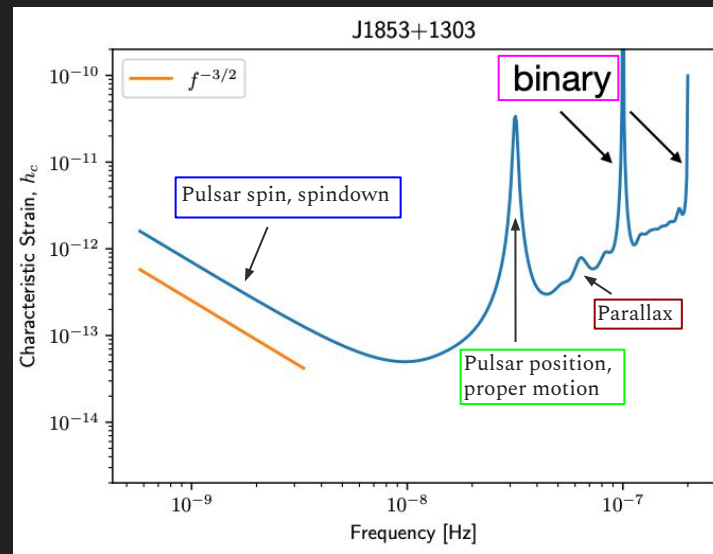
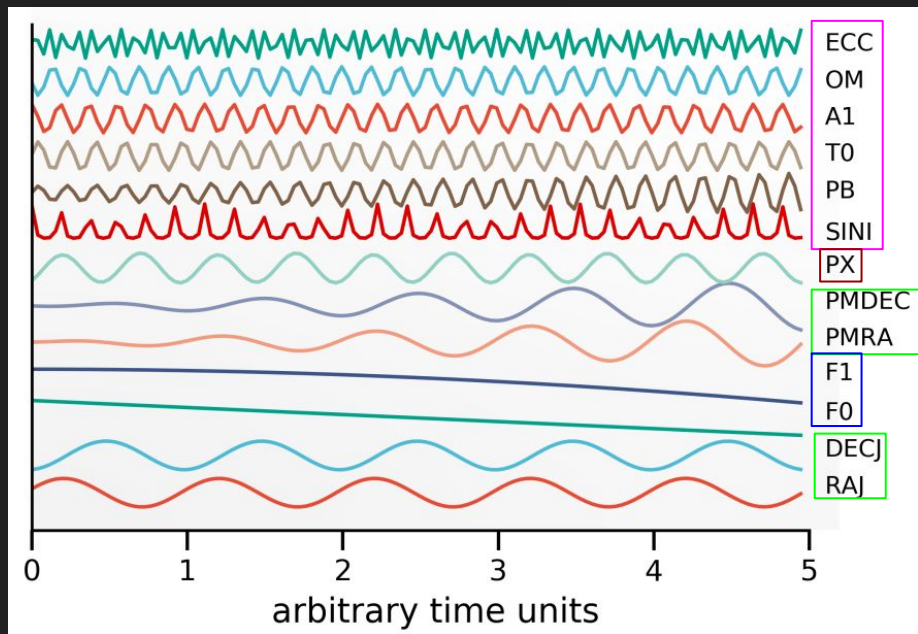
Bayesian noise analysis in practice



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Larsen, Baier+2026: Customized Chromatic Noise Models

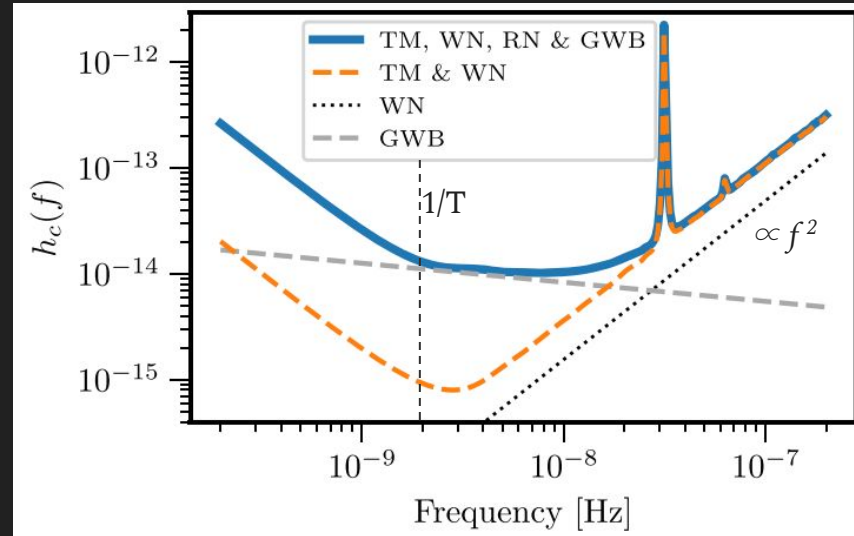
Timing model impacts on PTA sensitivity



Taylor, 2021: “The nHz gravitational wave astronomer”

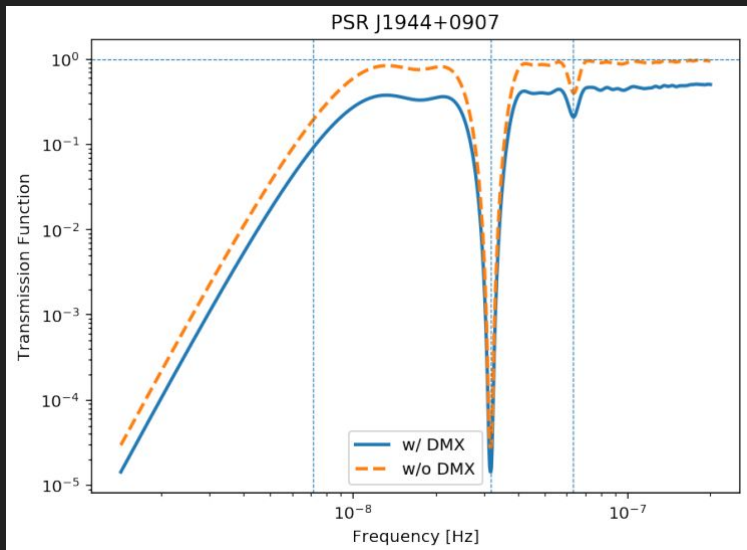
Hazboun, 2019: “Realistic Sensitivity Curves for PTAs”

Noise impacts on PTA sensitivity

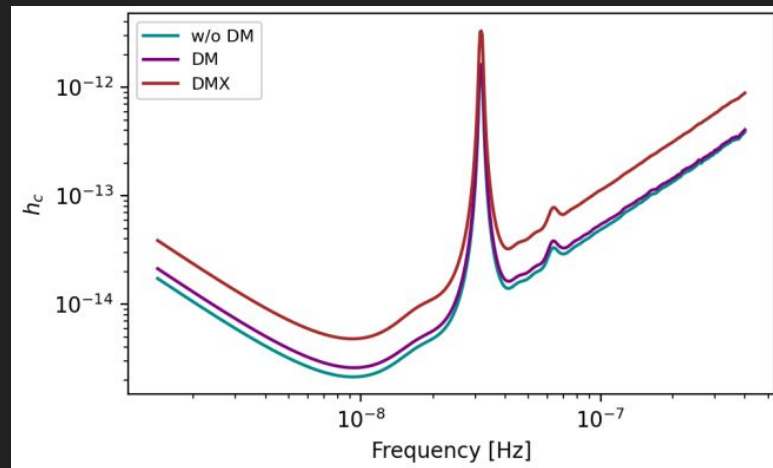


Agazie+2023: Detector Characterization and Noise Budget

DM modeling impacts on PTA sensitivity



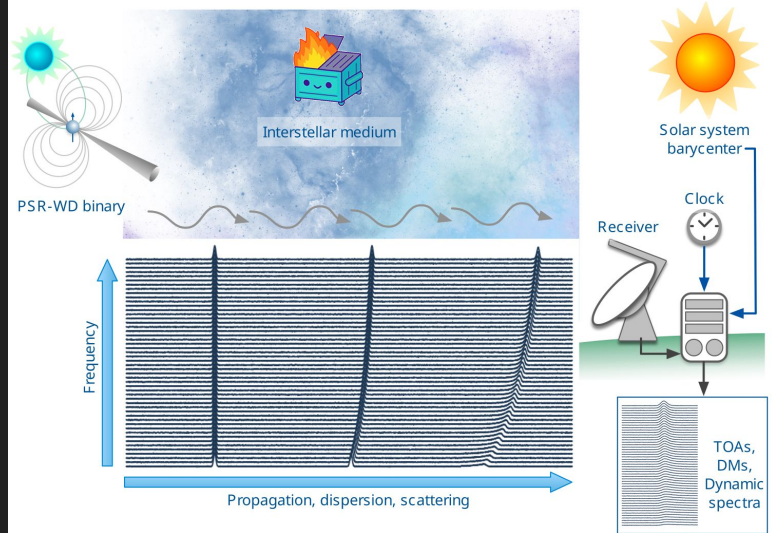
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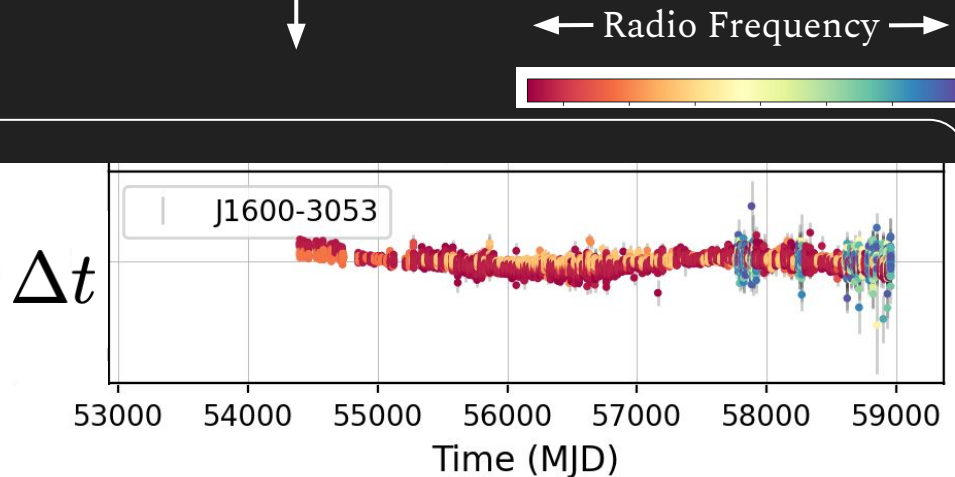
Gitika+2025: “Optimising the MeerKAT Pulsar Timing Array and towards precision pulsar timing with SKA-mid”

Data analysis problem

How to fit this...



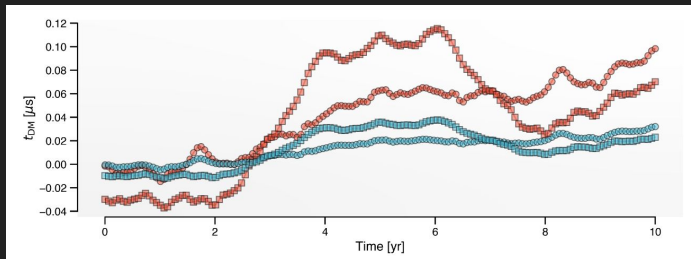
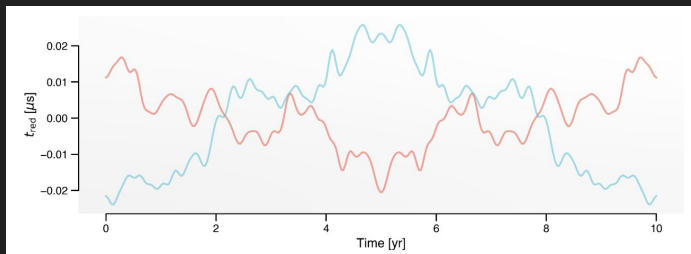
...to that?



Agazie+2023: “The NANOGrav 15 yr Data Set: Observations and Timing of 68 Millisecond Pulsars”

Data analysis problem

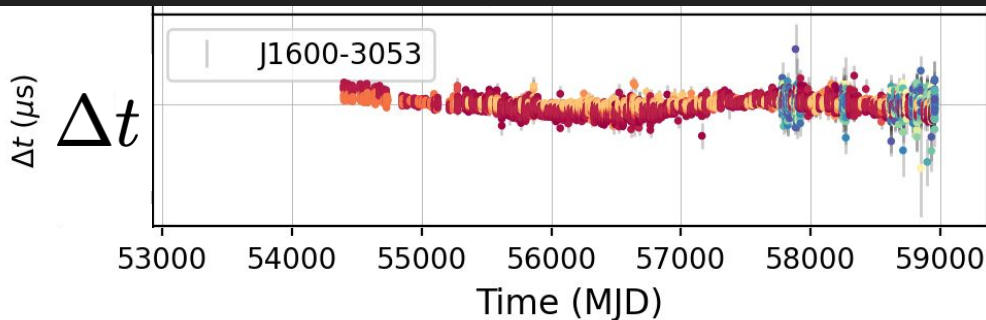
How to fit this...



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...to that?

← Radio Frequency →



Agazie+2023: “The NANOGrav 15 yr Data Set: Observations and Timing of 68 Millisecond Pulsars”

$\vec{\theta} \equiv$ parameters Likelihood Prior
 $d \equiv$ data

$$p(\vec{\theta}|d) = \frac{p(d|\vec{\theta})p(\vec{\theta})}{p(d)}$$

Posterior model evidence

$$y_i = \sum_j^{\# \text{ params}} \theta_j f_j(x_i)$$

parameters Basis functions

The linear model is a Gaussian process if $p(\vec{\theta})$ is Gaussian

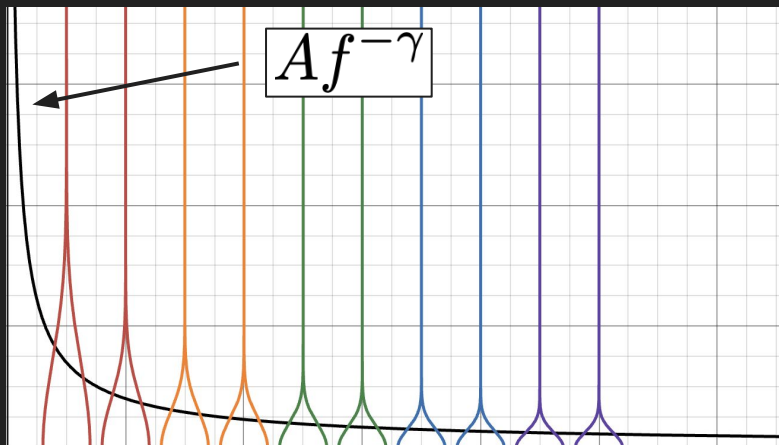
Key properties:

1. If likelihood is Gaussian, posterior is also Gaussian
2. We can analytically marginalize (integrate) over priors

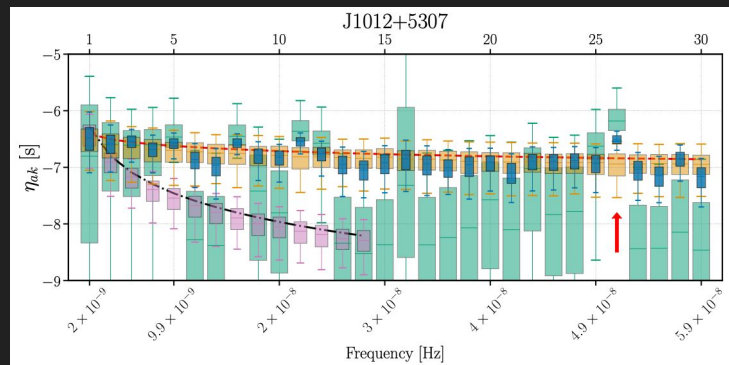
e.g. Fourier basis: $y_i = \sum_j^{N_f} [a_i] \cos(2\pi [f_j] t_i) + [b_i] \sin(2\pi [f_j] t_i)$

Prior: $p(a_i, a_j) = \mathcal{N}(0, \phi_{ij} \equiv \langle a_i a_j \rangle)$

Power law
red noise: $\phi_{ij} = \frac{A^2}{12\pi^3} \left(\frac{f}{f_{\text{ref}}} \right)^{-\gamma} \Delta f$



Posterior:



Agazie 2024: “The NANOGrav 15 yr data set: Posterior predictive checks for gravitational-wave detection with pulsar timing arrays”

Bonus slide: Equivalent perspectives on GPs

Full likelihood

$$p(\vec{\delta t} | \vec{a}, \vec{\theta}) = \frac{\exp \left(-\frac{1}{2} (\vec{\delta t} - \mathbf{F} \vec{a})^T \mathbf{N}^{-1} (\vec{\delta t} - \mathbf{F} \vec{a}) \right)}{\sqrt{2\pi \mathbf{N}}}$$

$$\boxed{\mathbf{C}(\vec{\theta}) = \mathbf{F} \phi \mathbf{F} + \mathbf{N}}$$

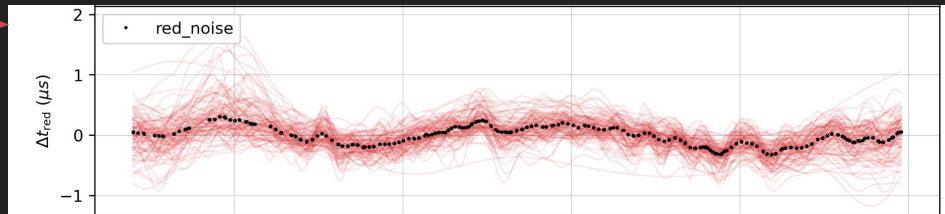
Marginalized likelihood

$$p(\vec{\delta t} | \vec{\theta}) = \frac{\exp \left(-\frac{1}{2} \vec{\delta t}^T \mathbf{C}^{-1} \vec{\delta t} \right)}{\sqrt{2\pi \mathbf{C}}}$$

Standard analysis of a single pulsar

Run MCMC to fit for:

- **Red Noise**
 - e.g. pulsar spin fluctuations + GWs
 - Basis functions: Fourier series
 - Sample 2 hyperparameters (A , γ)



Standard analysis of a single pulsar

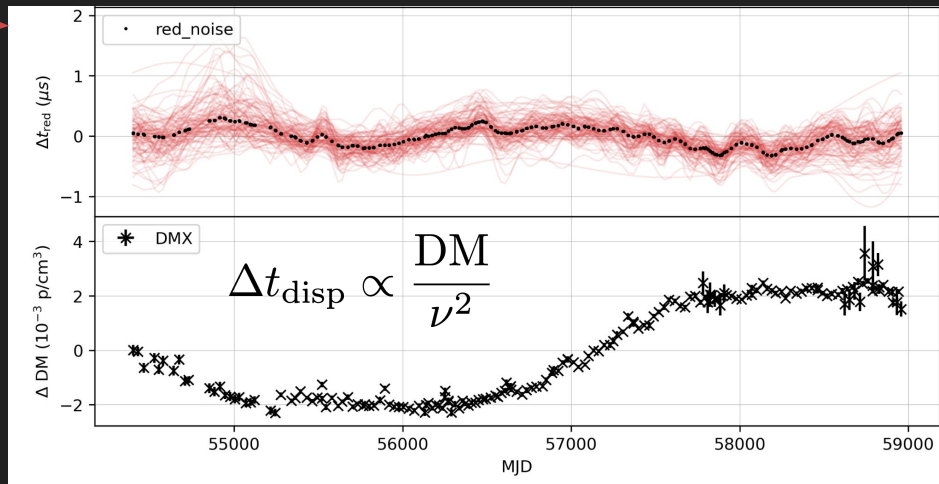
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- **Dispersion Measure (DM) model (DMX)**

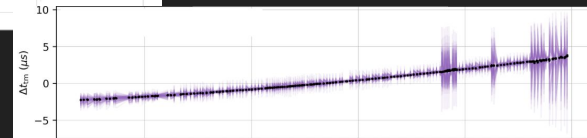
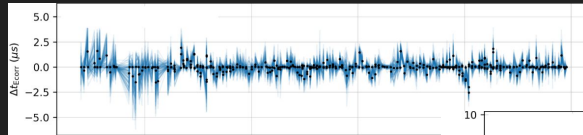
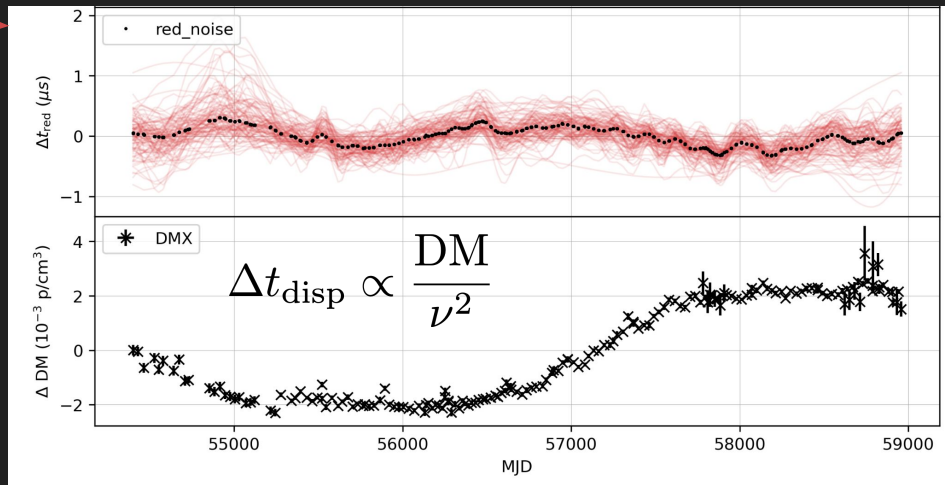
- Basis functions: Piecewise constants
- Sample 0 hyperparameters in MCMC
- 100s of marginalized params



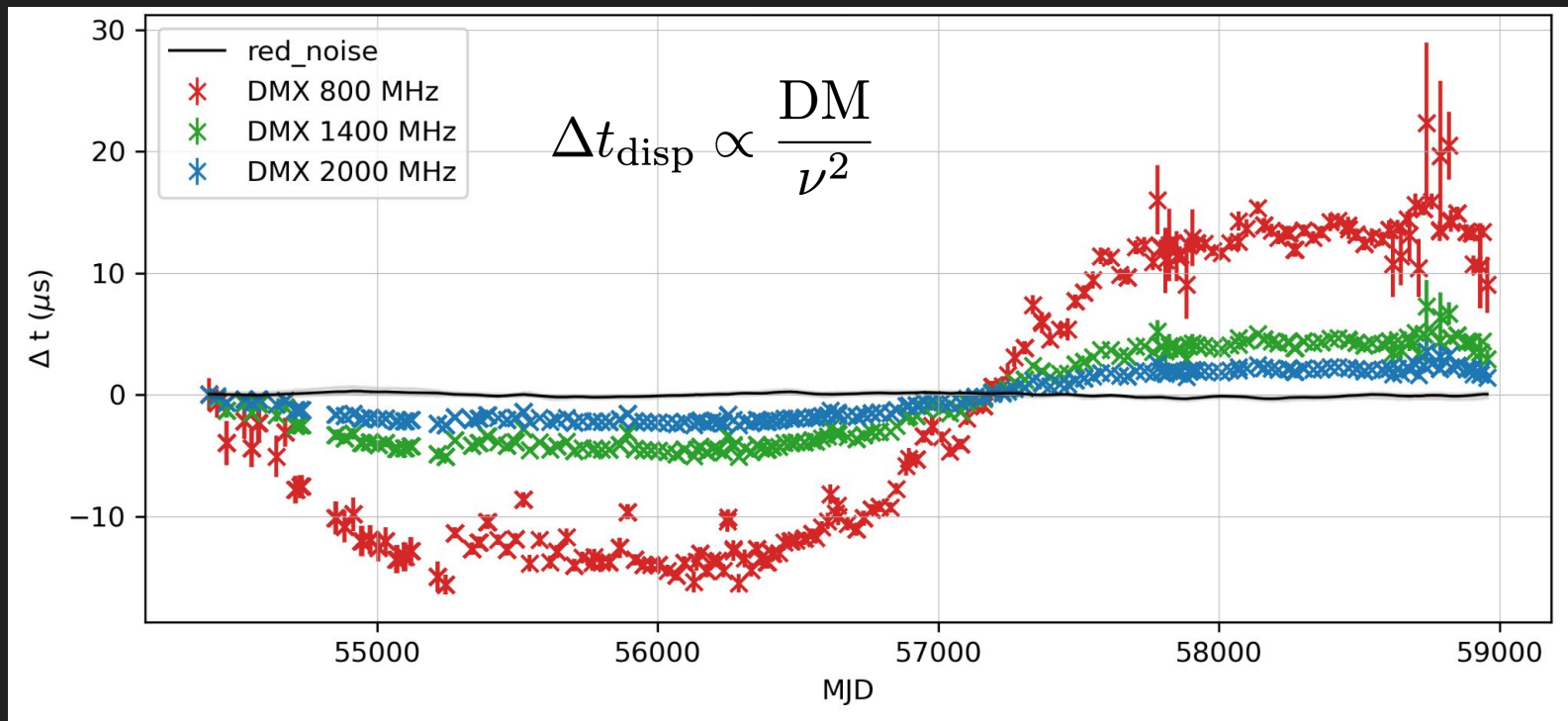
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- **Dispersion Measure (DM) model (DMX)**
 - Basis functions: Piecewise constants
 - Sample 0 hyperparameters in MCMC
 - 100s of marginalized params
- Also **white noise** and remaining **timing model variations!**



Time-domain representation: J1600-3053

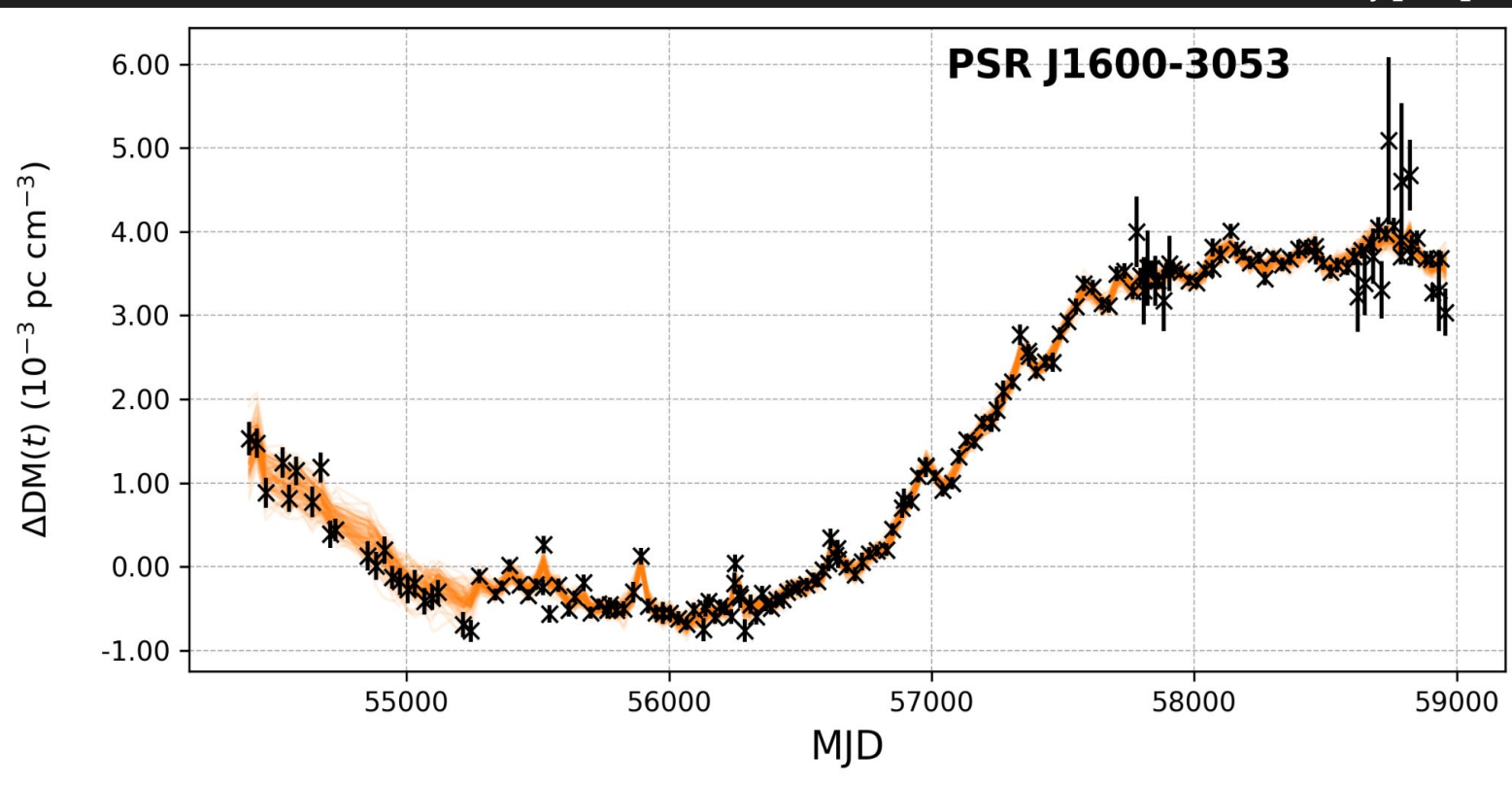


Chromatic noise can be much louder than achromatic red noise!
→ Suggests testing different models for DM

Alternative model part 1: Use smooth basis functions (DM red noise)

(Lentati+2016, Lam+2018, Hazboun+2020, Goncharov+2021, Chalumeau+2022, Larsen+2024)

Comparing DM red noise vs DMX (Now sampling a model w/
1-9 hyperparameters)



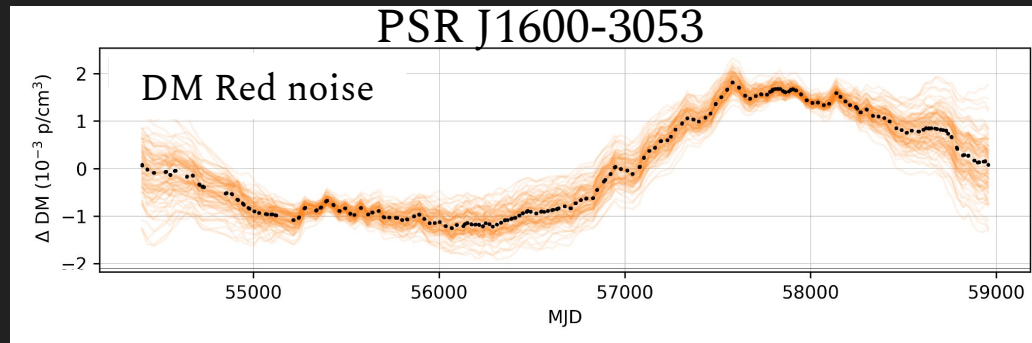
Alternative noise model: Use a time-correlated DM model (Gaussian process)

(Lentati+2014, Chalumeau+2022)

$$\Delta t_{\text{disp}} \propto \frac{\text{DM}}{\nu^2}$$

- **DM Red noise**

- e.g. ISM turbulence ($\gamma = 8/3$)
- Power law spectrum



Alternative noise model: Use a time-correlated DM model (Gaussian process)

(Lentati+2014, Chalumeau+2022)

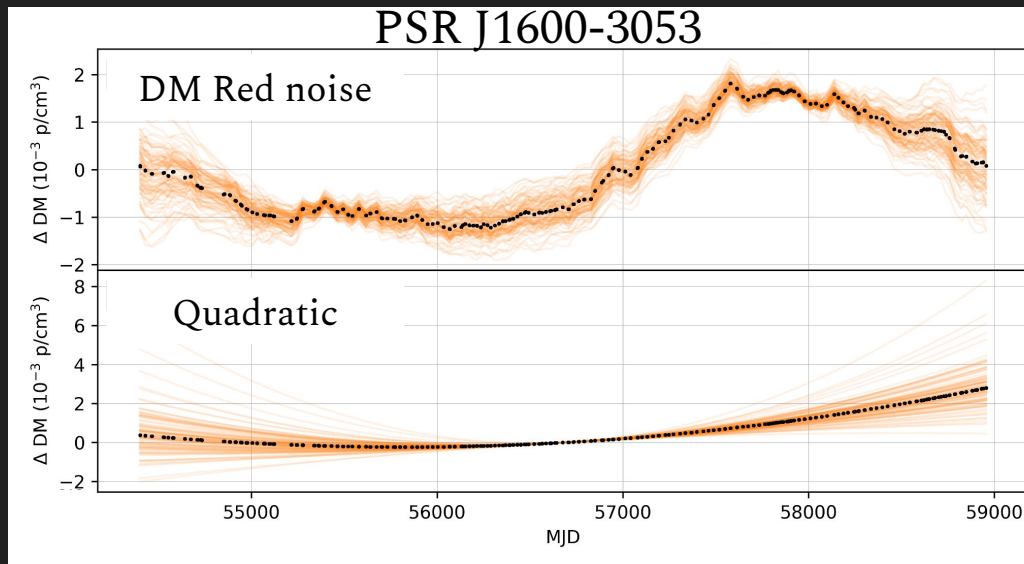
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- **DM Red noise** →

- e.g. ISM turbulence ($\gamma = 8/3$)
- Power law spectrum

- **Quadratic trend** →

- e.g. changing distance to pulsar
- Deterministic (DM, DM1, DM2)



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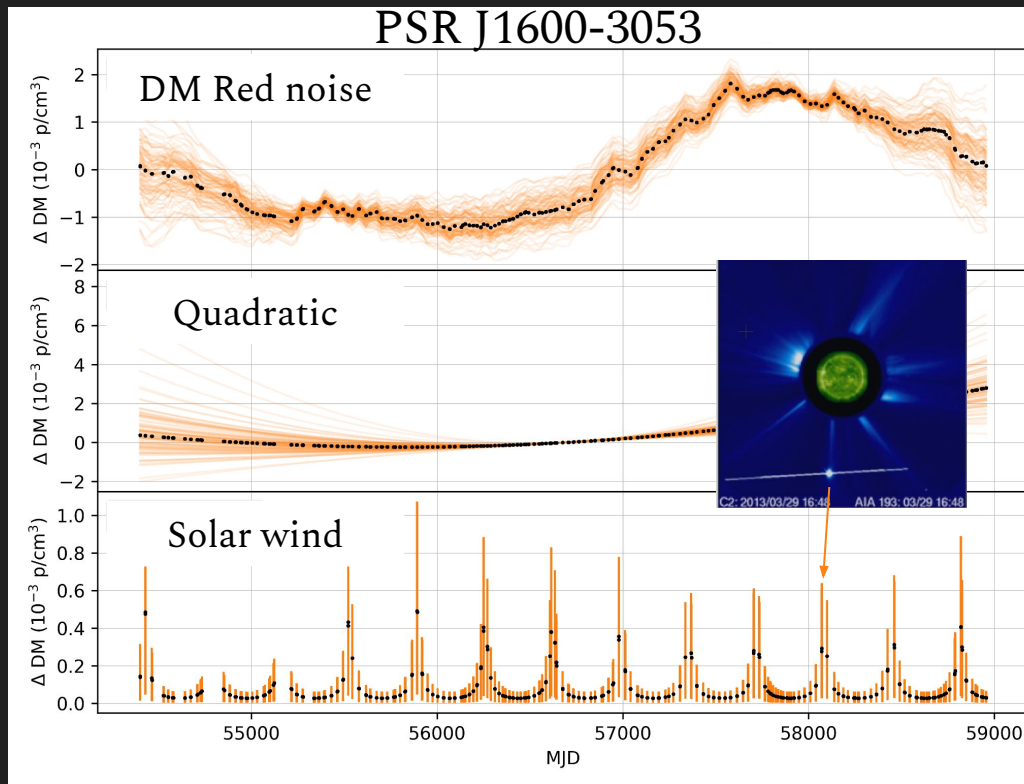
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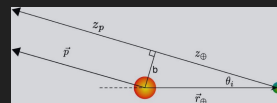
- e.g. changing distance to pulsar
- Deterministic (DM, DM1, DM2)

- **Solar wind model**
(Hazboun+2022)

- Annual changes to DM
- Assuming a symmetric solar wind
- Deterministic (NE_SW)
- ...or stochastic (more on this later)
- Ecliptic latitude-dependent



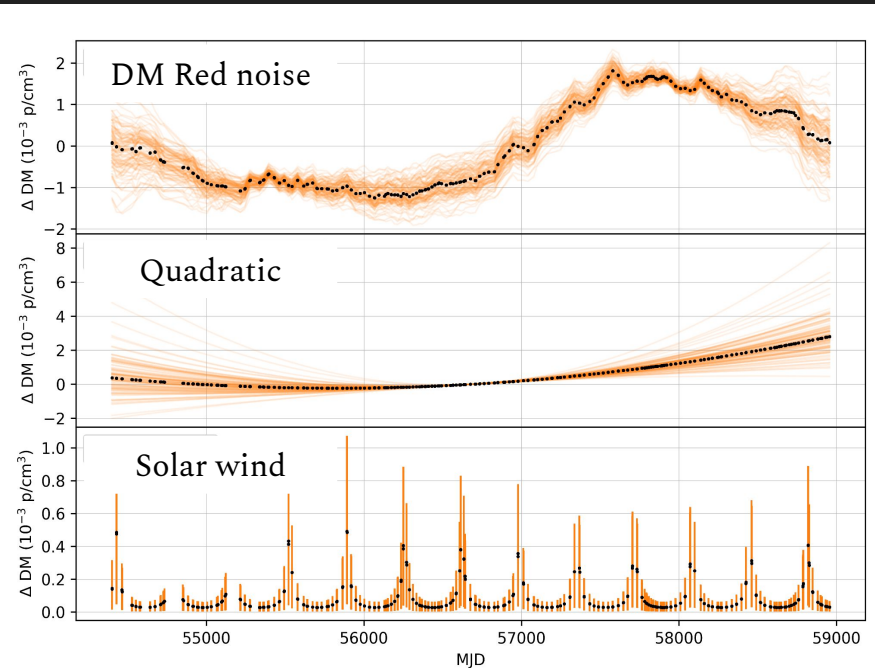
$$n_e(\mathbf{r}) = N_e^{\text{SW}} \left(\frac{1 \text{ au}}{|\mathbf{r}|} \right)^2$$



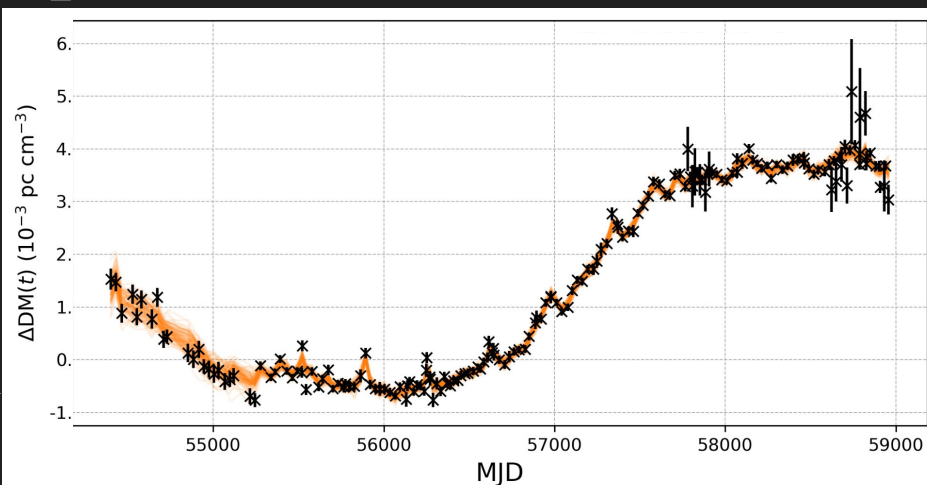
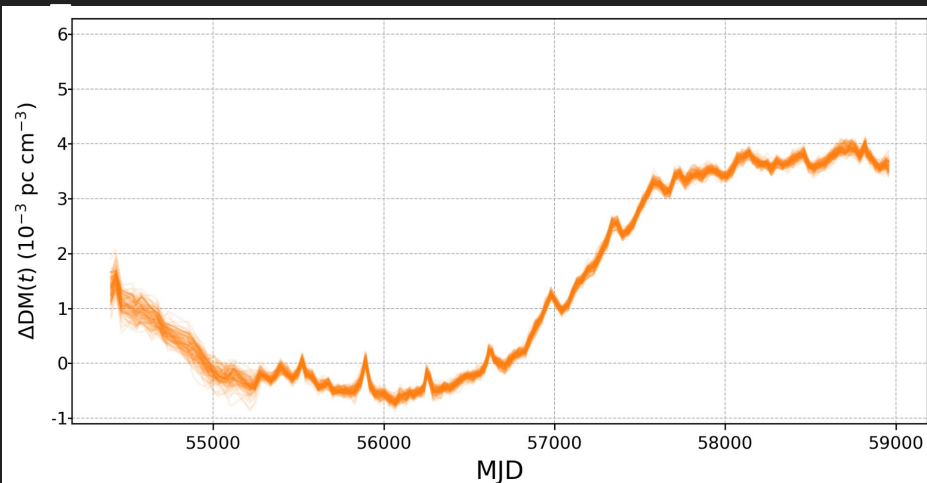
Alternative noise model

(Lentati+2014, Chalumeau+2022)

$$\Delta t_{\text{disp}} \propto \frac{\text{DM}}{\nu^2}$$

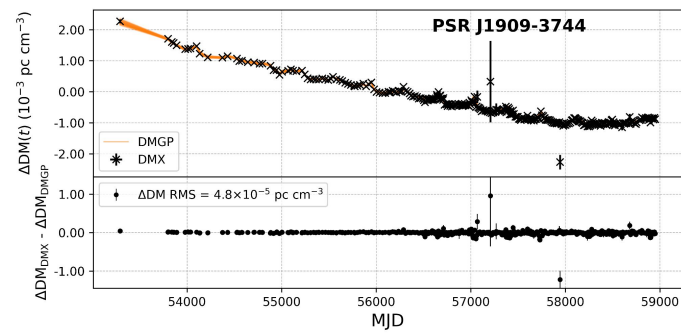
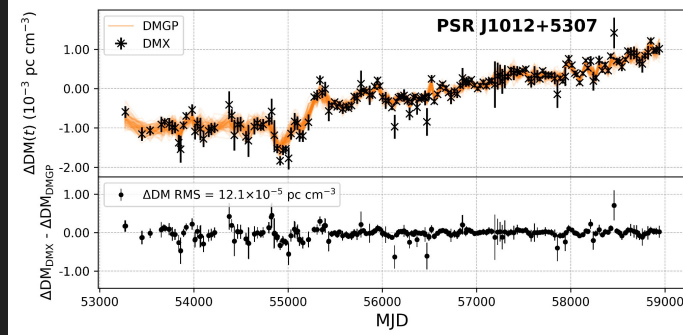
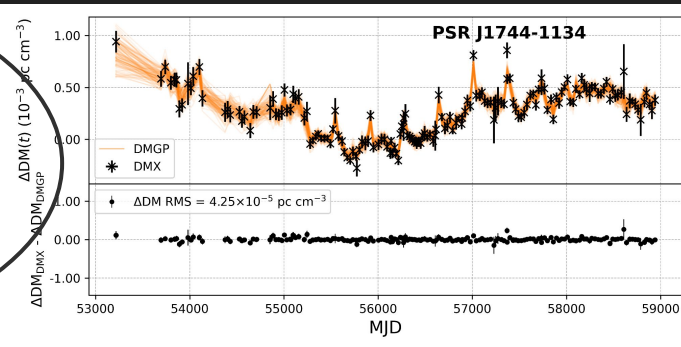
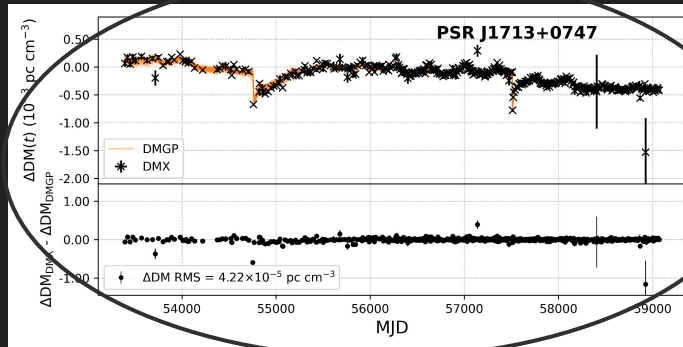
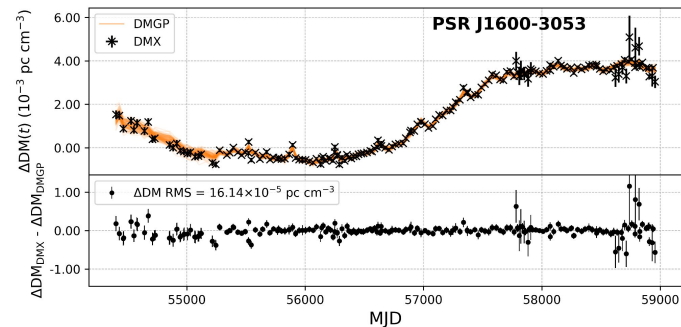
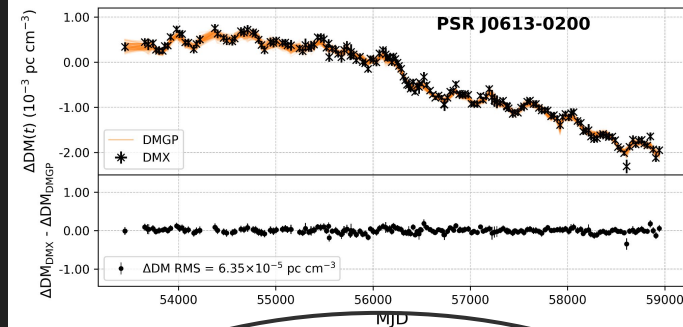


Compare to DMX model



Comparing DM fits: 6 pulsars

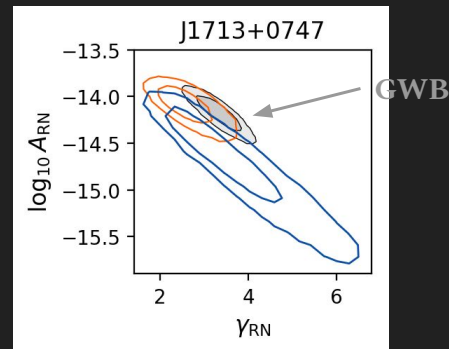
- Time-correlated DM model can reproduce DMX series (Lentati+2017, Larsen+2024, Iraci+2024)
- Variety of different trends



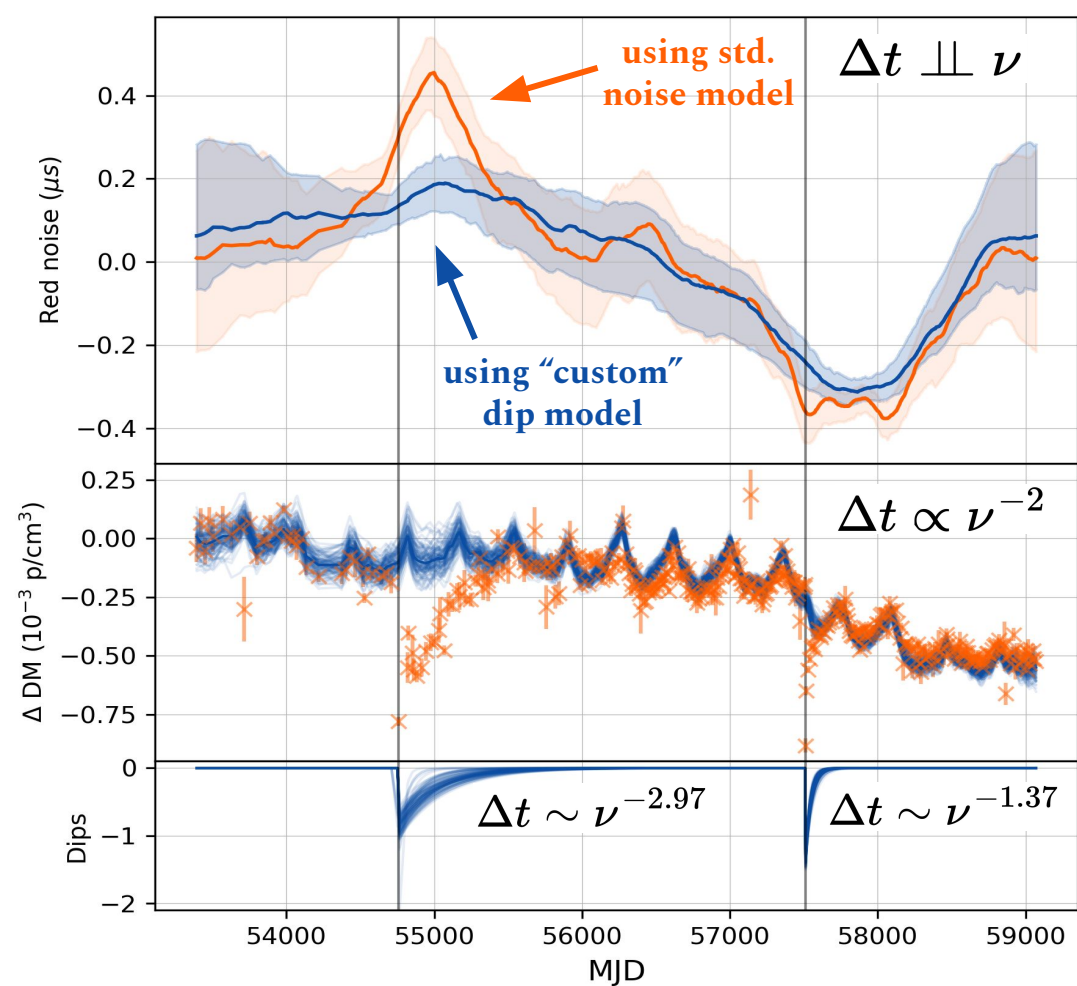
Is a DM-only model good enough?

Try fitting out events using a deterministic “dip” model

$$s^{\text{exp}}(t_i) = -A_{\text{exp}} \Theta(t_i - t_{0,\text{exp}}) \exp\left(-\frac{t_i - t_{0,\text{exp}}}{\tau_{\text{exp}}}\right) \times \left(\frac{\nu_i}{1400 \text{ MHz}}\right)^{-\chi_{\text{exp}}}$$



Accounting for this drives down the amplitude of the GWB

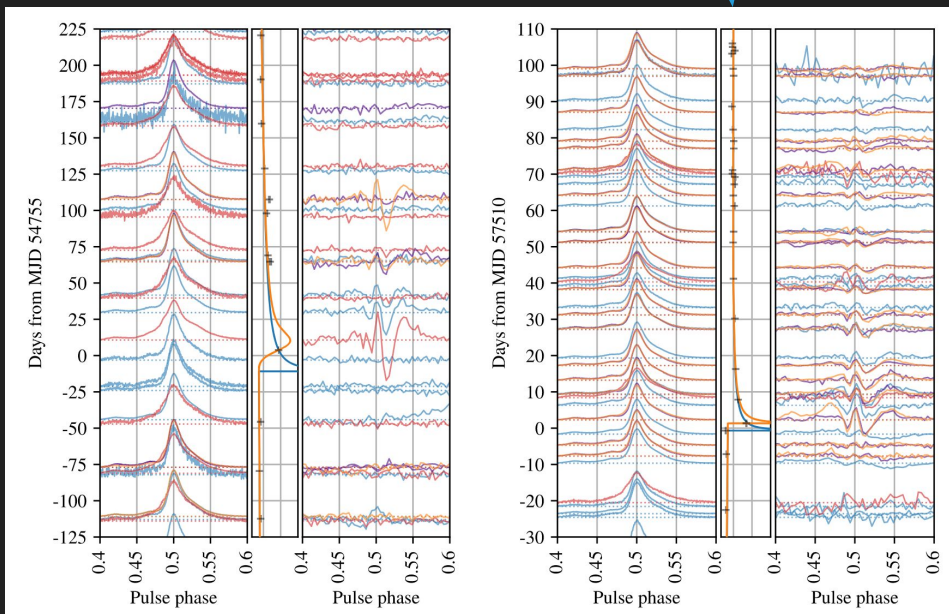
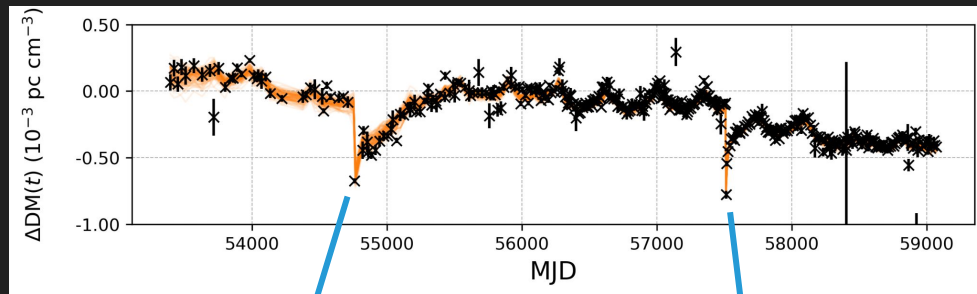


Addition to model: PSR J1713+0747 *chromatic events*

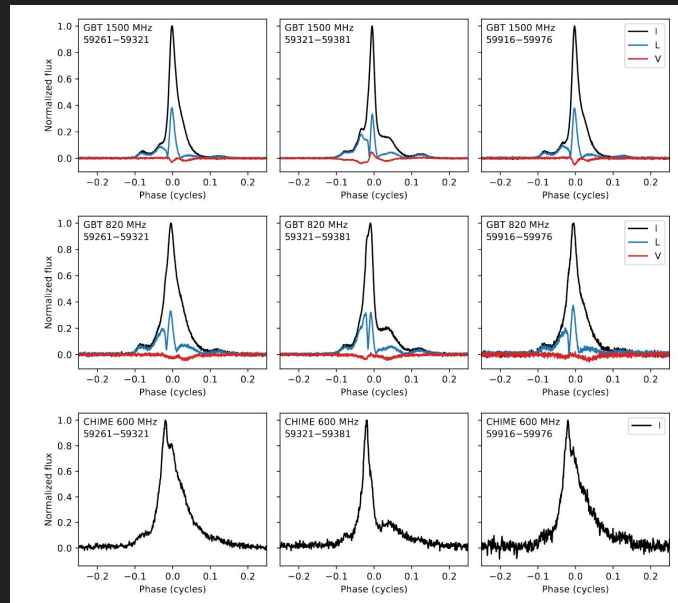
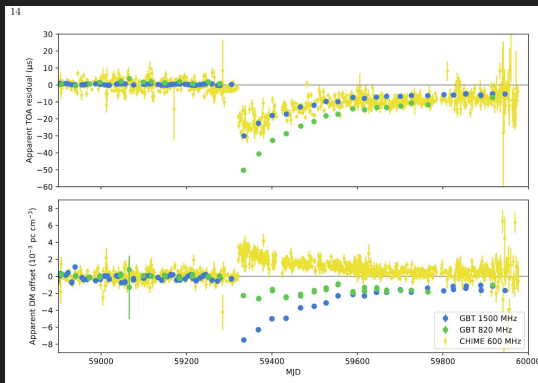
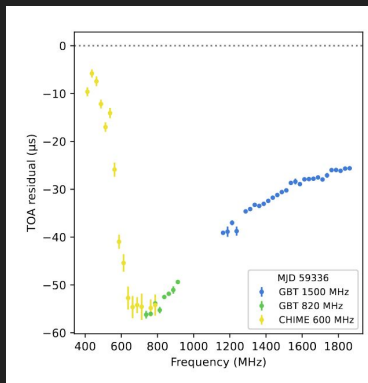
- Changes to the *pulse profile* at the times of these events (Lin+2021)
- These do not follow

$$\Delta t_{\text{disp}} \propto \frac{\text{DM}}{\nu^2}$$

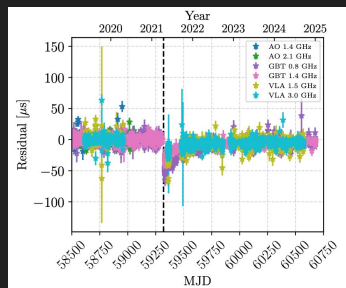
(Lam+2018, Goncharov+2021, Chalumeau+2022)



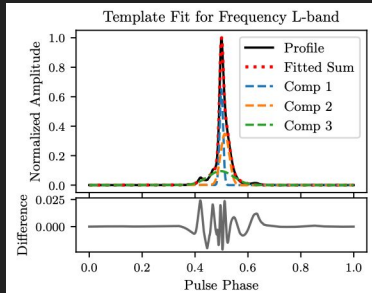
Recent dramatic profile change in J1713+0747



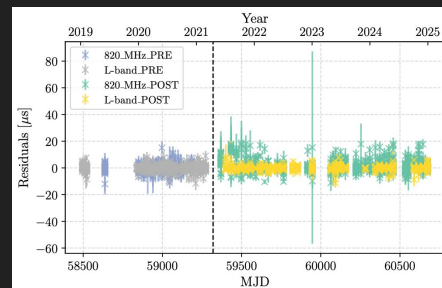
Jennings+2022, Mandow+2025



Switch to
pulse profile
domain

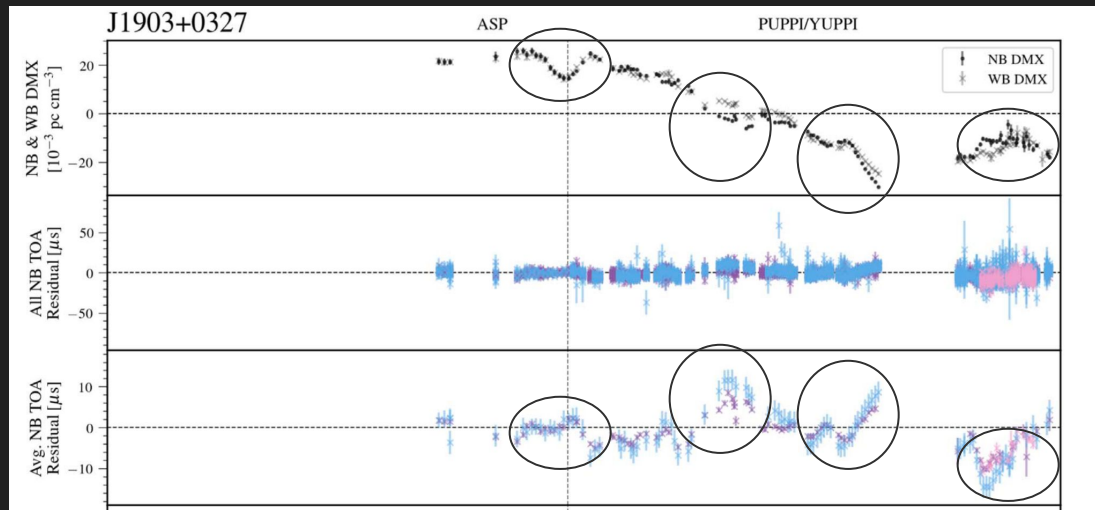


Regenerate
TOAs



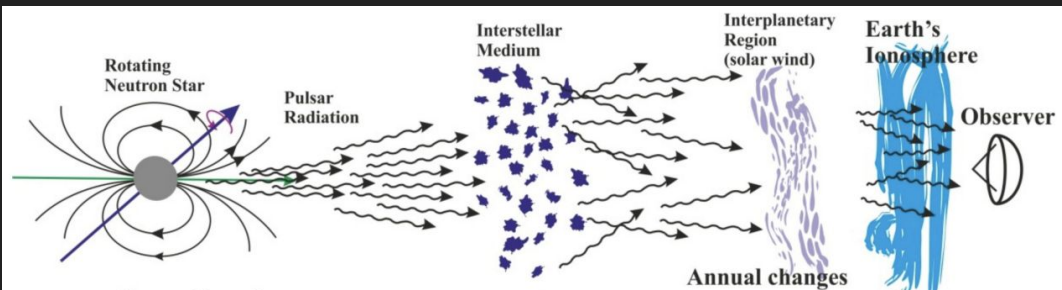
Nichols+2026

What about this?

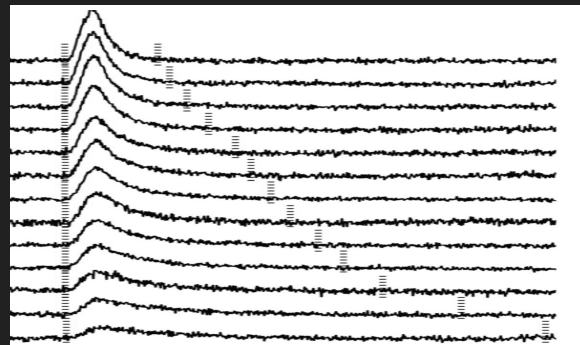


Agazie+2023: “The NANOGrav 15 yr Data Set:
Observations and Timing of 68 Millisecond Pulsars”

Scattering noise (non-dispersive)



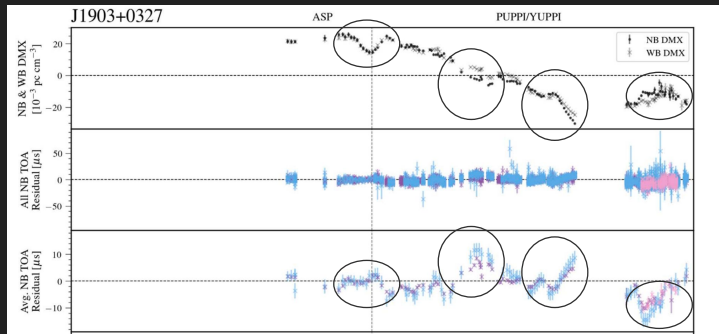
Blaszkiewicz et al. 2020



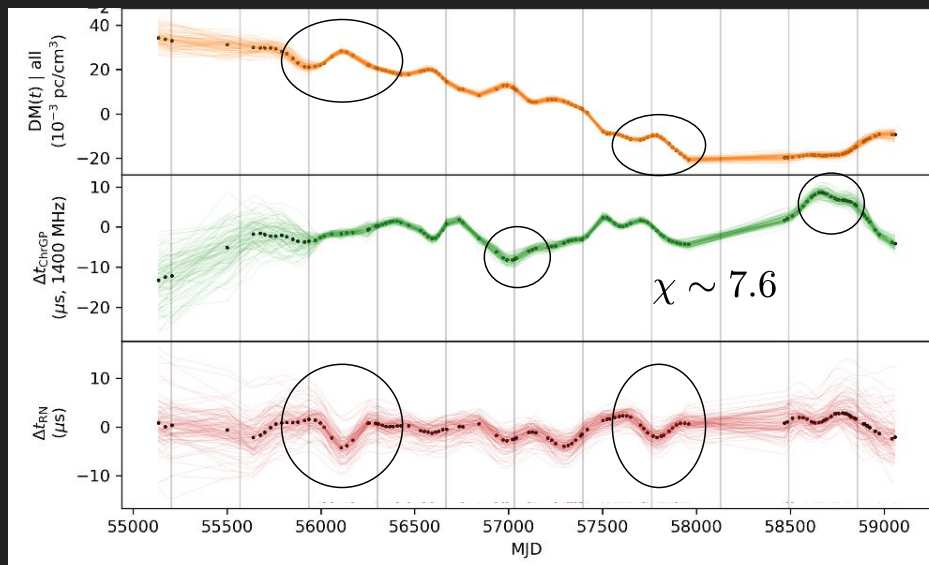
Condon and Ransom 2016

Prediction from ISM models
(under many assumptions): $\Delta t_{\text{scat}} \sim \nu^{-4}$

Attempted fix: Add an extra GP with $\Delta t \propto \nu^{-x}$



Attempted fix: Add an extra GP with $\Delta t \propto \nu^{-\chi}$



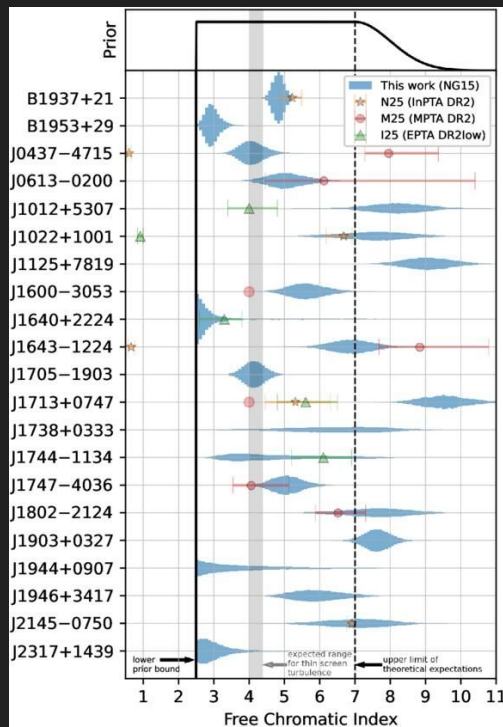
$$\Delta t \propto \nu^{-2}$$

$$\Delta t \propto \nu^{-\chi}$$

$$\Delta t \perp \nu$$

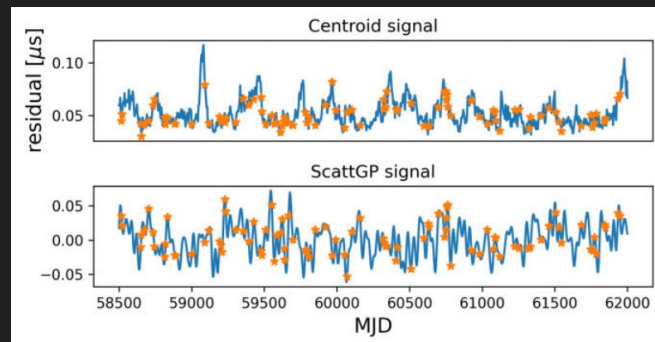
Larsen, Baier+2026: Customized
Chromatic Noise Models

Issues with current scattering noise models

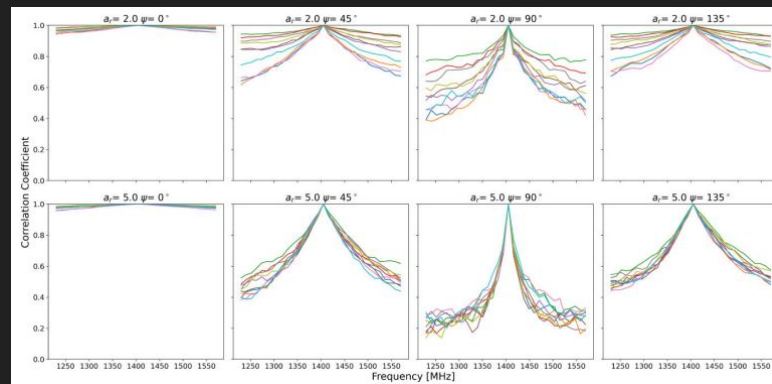


Larsen, Baier+2026: Customized Chromatic Noise Models

Discrepancies in scattering models between PTAs



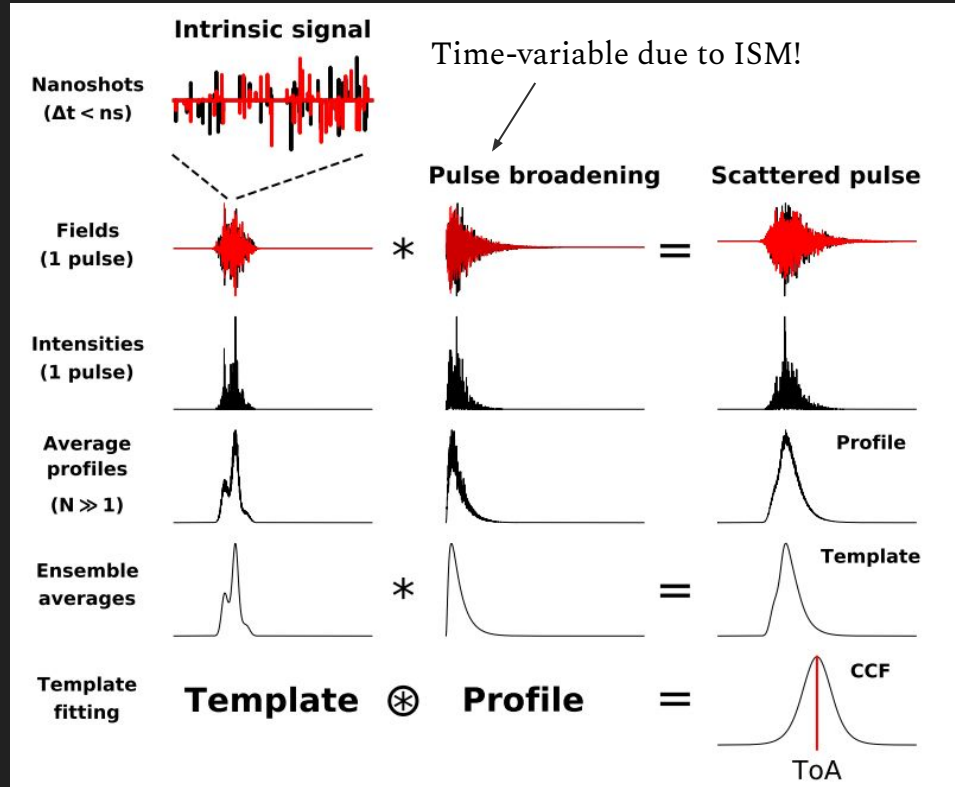
Kulkarni+2025:
Measuring scattering variations in pulsar timing observations:
a test of the fidelity
of current methods



Future scattering noise models require:

- Accounting for non-Gaussianity
- Accounting for decorrelations over frequency

Promising mitigation techniques in the profile domain in development



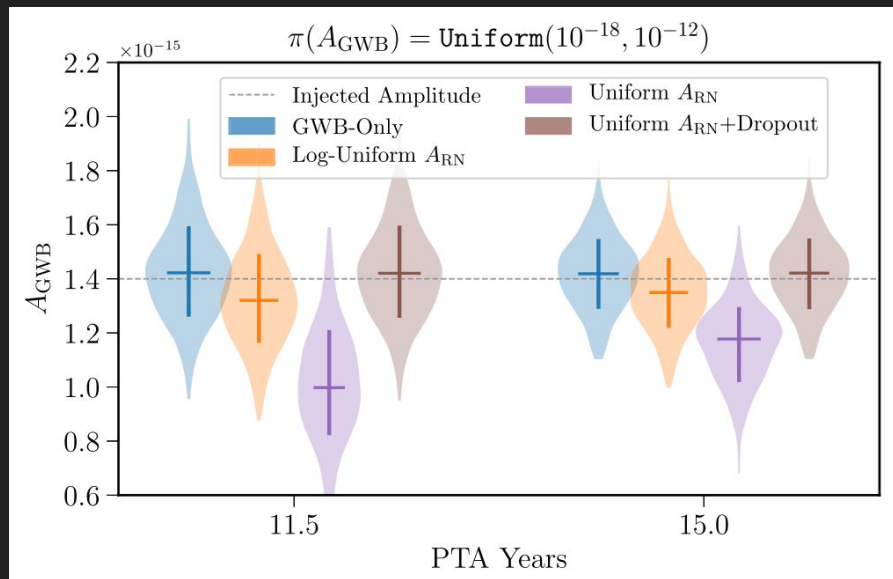
- Cyclic spectroscopy (Demorest+2011, Dolch+2021, Turner+2023, Turner+2024)
- CLEAN deconvolution algorithm (Young and Lam 2023)

Bonus slide: Always include your intrinsic red noise

- GWB is a red noise process
- Additional red noise w/ appropriate priors must be included in individual pulsars for robust inference
- Std in field: Uniform in log-amplitude or *hierarchical*

Refs:

- Hazboun+2020: Model Dependence of Bayesian Gravitational-wave Background Statistics for PTAs
- van Haasteren 2025: Use model averaging instead of model selection in pulsar timing



Hazboun+2020

How I learned to stop worrying and love custom pulsar noise model selection!

NANOGrav 12.5yr and 15 yr
custom noise analysis
(Hazboun+2026;
Larsen, Baier+2026)

Key takeaways from model selection:

1. Must consider basis size
2. Models are highly sensitive to variations in the solar wind
3. Need to make models faster

Other recent works:

- Reardon+2023 (PPTA noise models)
- Goncharov+2024 (EPTA noise models)
- Susarla+2025 (LOFAR solar wind models)
- Nobleson+2026 (InPTA noise models)
- Chen+2026 (CPTA noise models)



Covering our bases

$$N_{\text{freqs}} \in [50, 100, 150, 200]$$

Basis size selection is critical for proper noise characterization

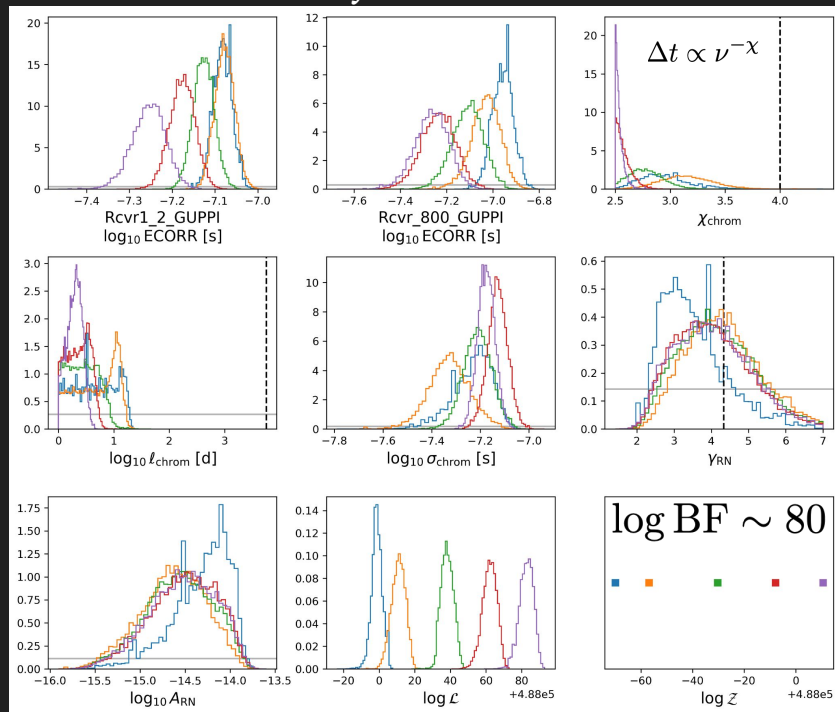
Disjoint posteriors:

- Posterior reweighting ✗
- HyperModel ✗
- Nested Sampling ✓

Other options:

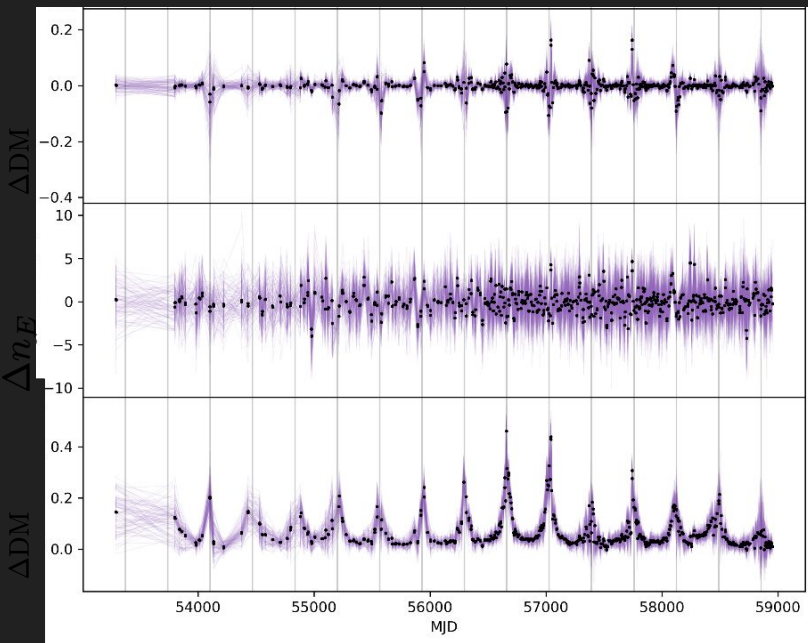
- Basis sampling (EPTA et al)
- NG20: Bayesian Blocks (Thompson, Stairs in prep.)

PSR J1909+3744



Perturbations to the deterministic solar wind model

PSR J1909+3744

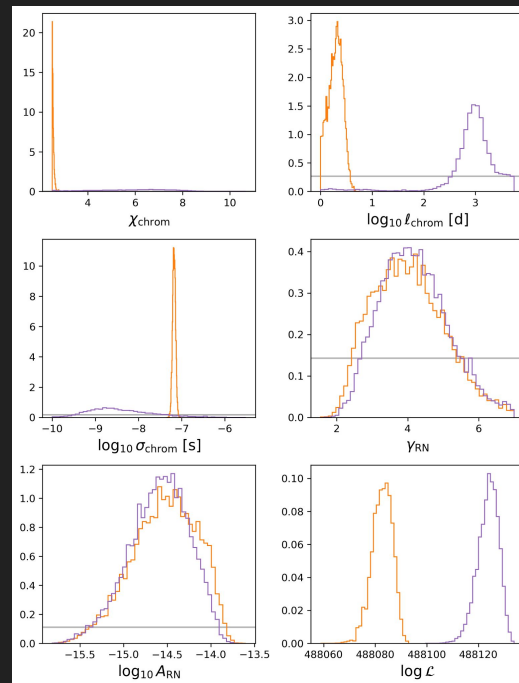


Larsen, Baier+2026: Customized Chromatic Noise Models

SW GP
perturbations

Rescaled to
electron density

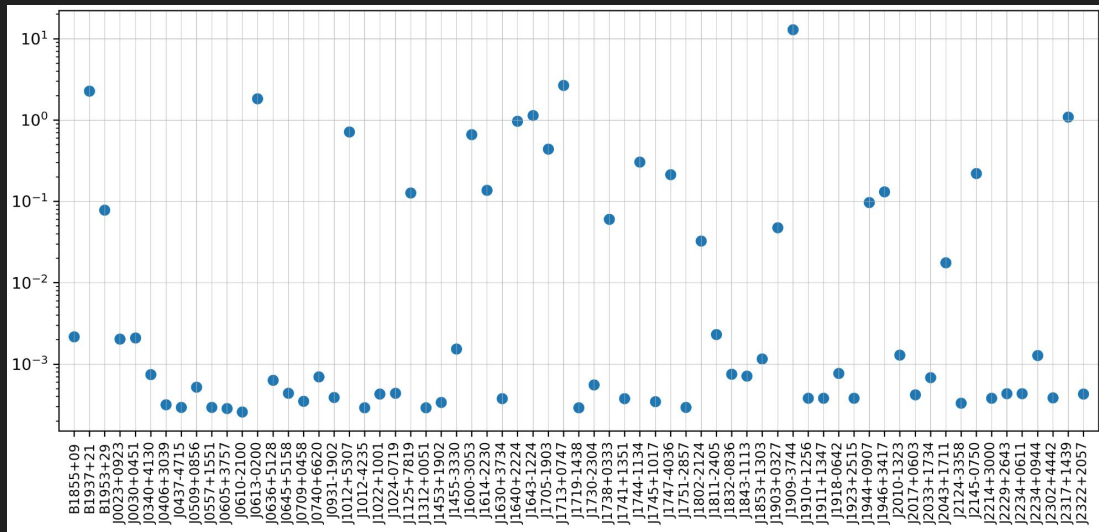
**Deterministic
model + SWGP**



Short timescale SW variations may masquerade as scattering noise

These models could be faster

Single Likelihood Evaluation Times (s)

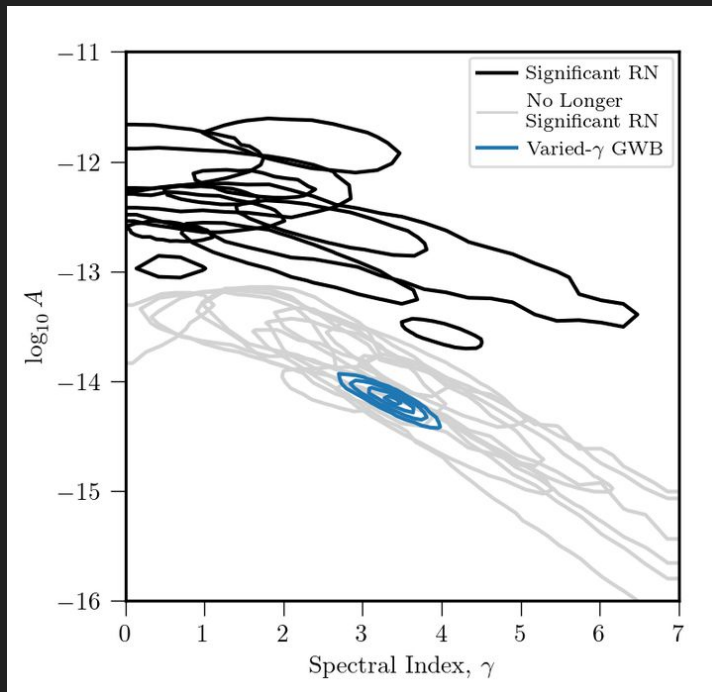


Possible avenues for improvement:

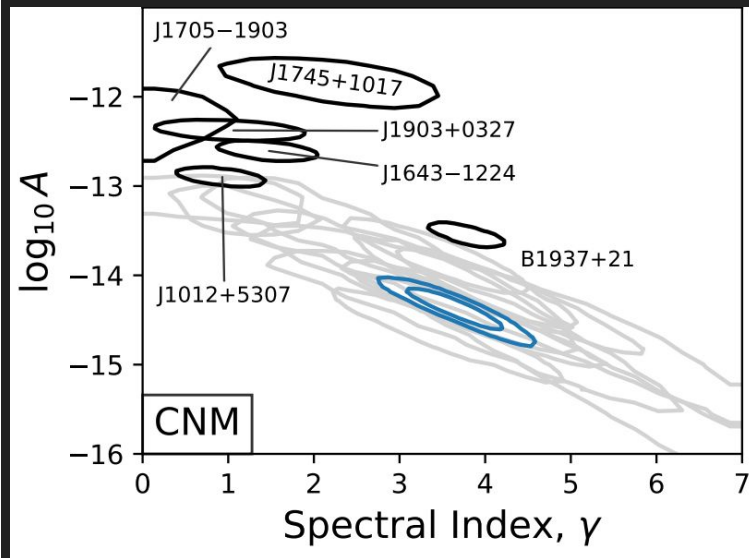
- Discovery
- Prometheus
- Smarter GP basis implementations
- ...?

Results 1/3: Change to noise across the PTA

Using DMX



Using customized chromatic models



Results 2/3: GWB results

Optimized models prefer steeper GWB spectrum

standard

$$\gamma_{CRN} = 3.35^{+0.33}_{-0.32}$$
$$\log_{10} A_{CRN} = -14.17^{+0.13}_{-0.12}$$

custom chromatic

$$\gamma_{CRN} = 3.67^{+0.37}_{-0.38}$$
$$\log_{10} A_{CRN} = -14.37^{+0.17}_{-0.15}$$

9% increase in gamma. 38% decrease in A

Optimized models increase HD significance

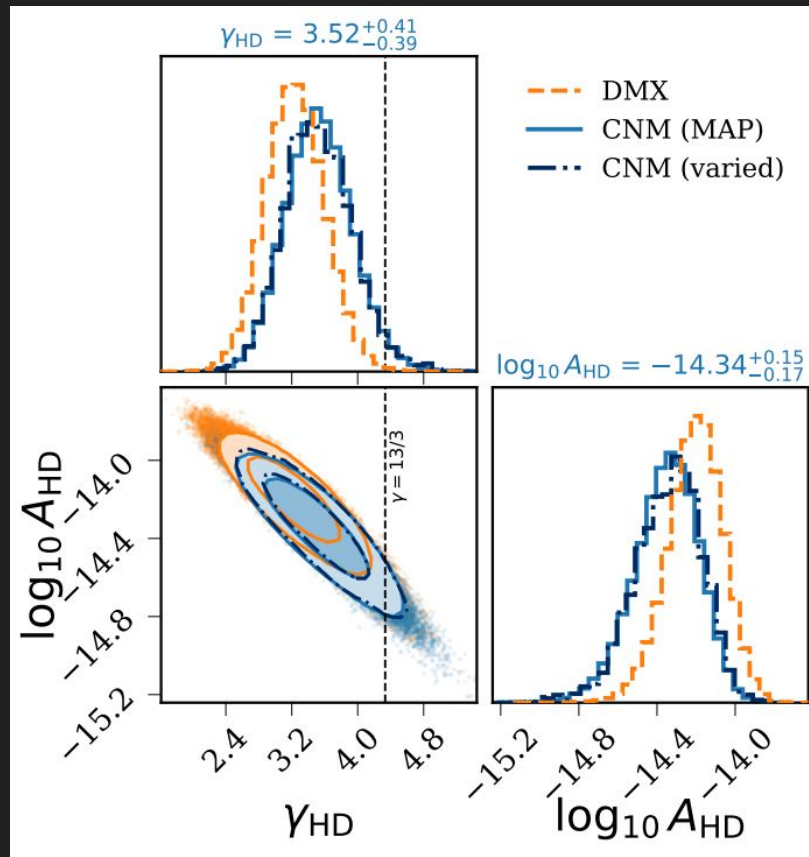
standard

$$B_{CRN}^{HD} = 226 \pm 70$$

custom chromatic

$$B_{CRN}^{HD} = 1530 \pm 124$$

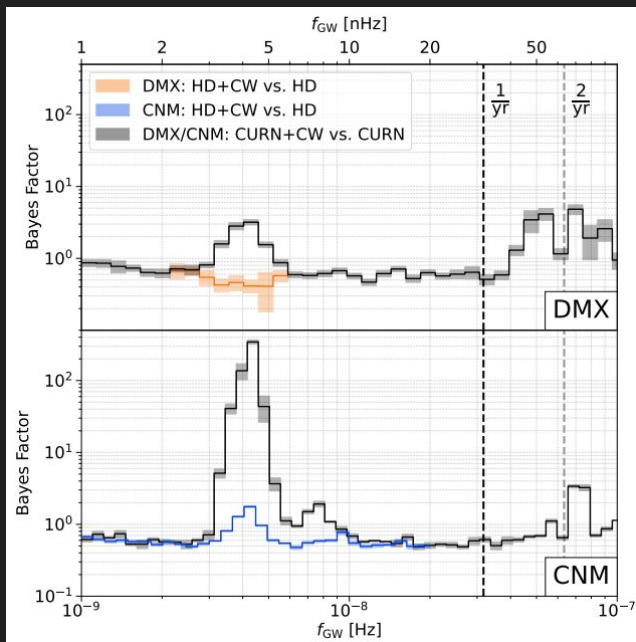
via resampling: Hourihane et al. 2023



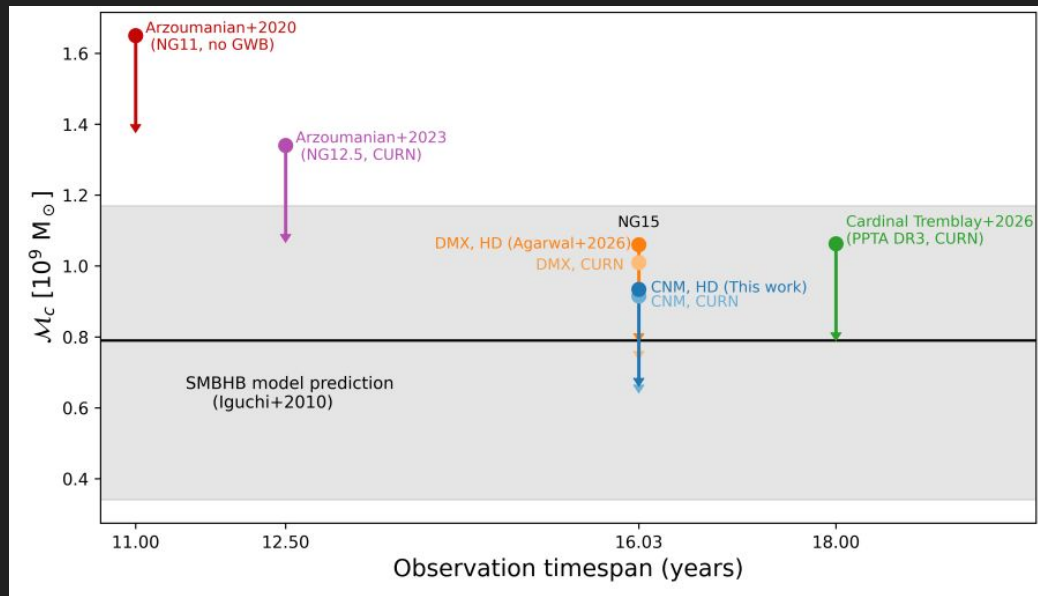
Agarwal+2026: Impacts of Customized Chromatic Noise Models

Results 3/3: CW results

Agarwal+2026: Impacts of Customized Chromatic Noise Models



Increased detection
significance for CW candidate
at 4 nHz



Improved strain/chirp mass
upper limit on 3C66B

Better noise modeling leads to
improved GW measurements



There are always more ways to improve
our noise models



Thanks!

Reach out with any questions/requests for further reading bjorn.larsen@nanograv.org